Natural Language Understanding: Assignment 2

Final Report on Word Embeddings and Character-Level Sequence Models

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1 Problem 1: Learning Word Embeddings from IIT Jodhpur Data

1.1 Dataset Preparation and Corpus Statistics

Textual data was scraped and compiled from various IIT Jodhpur sources, including official website pages, academic regulations, and course syllabi. The raw data underwent preprocessing to remove boilerplate formatting, followed by tokenization, lowercasing, and the removal of excessive punctuation to construct a clean English corpus.

The final preprocessed dataset exhibits the following statistics:

- **Total Documents:** 22
- **Total Sentences:** 1966
- **Total Tokens:** 53,496
- **Vocabulary Size:** 5,376
- **Average Tokens per Sentence:** 27.2
- **Total Corpus Size:** 0.351954 MB

Top-10 Most Frequent Words:

1. `the` (2576)
2. `of` (2278)
3. `and` (1637)
4. `to` (1137)
5. `in` (1038)
6. `for` (1009)
7. `on` (459)
8. `by` (371)
9. `be` (360)
10. `is` (358)

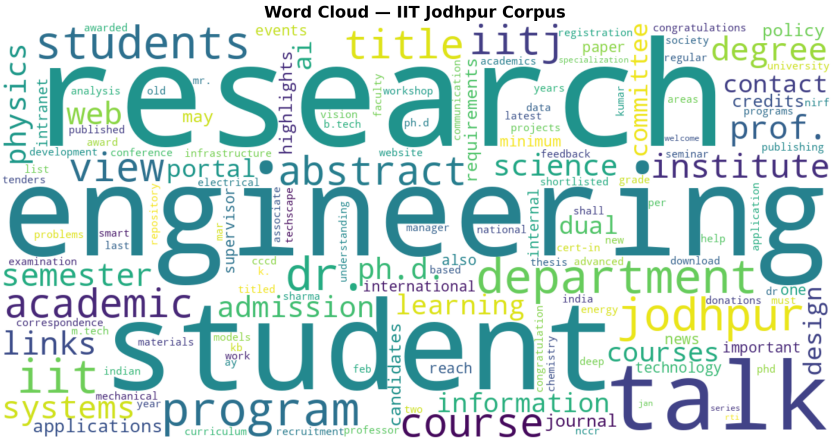

Figure 1: Word Cloud illustrating the most frequent words in the IIT Jodhpur corpus.

1.2 Model Training and Hyperparameter Search

Two Word2Vec architectures, Continuous Bag of Words (CBOW) and Skip-gram with Negative Sampling, were trained from scratch. We formalized the grid-search evaluation by averaging model performance (Analogy Solved Score and Silhouette Clustering Score) over all hyperparameters: (i) Embedding Dimension, (ii) Context Window Size, and (iii) Number of Negative Samples.

The averaged impacts of these hyperparameter adjustments are documented in Table 1.

Table 1: Averaged Hyperparameter Search Results

(a) Model Architecture

Model Type	**Avg Analogy**	**Avg Silhouette**
CBOW	0.2532	-0.00718
Skip-gram	0.1646	0.00044

(b) Effect of Embedding Dimension

Dimension	**Avg Analogy**	**Avg Silhouette**
50	0.3333	-0.00478
100	0.2105	-0.00556
200	0.1842	-0.00425
300	0.1875	-0.00464
350	0.1875	0.00235

(c) Effect of Context Window Size

Window Size	**Avg Analogy**	**Avg Silhouette**
3	0.3158	0.00359
5	0.1579	-0.00483
7	0.3889	-0.00907
8	0.1250	-0.00586
10	0.1250	-0.00419

(d) Effect of Negative Samples

Neg Samples	**Avg Analogy**	**Avg Silhouette**
5	0.4048	-0.00438
10	0.1905	-0.00364
15	0.1667	-0.00500
20	0.0312	0.00046

Best Model Configuration Highlights:

- **Best Model for Analogies:** `skipgram_d300_w8_n20`
- **Best Model for Semantic Clustering (Silhouette Score = 0.0121):** `skipgram_d200_w7_n10`. Skip-gram models consistently produced higher structural cluster definition than their CBOW counterparts.

1.3 Semantic Analysis

The learned vector representations successfully capture semantic meaning within the academic domain. For instance, evaluating the 300-dimensional vector for the word *"student"* yields the following leading values:

```
[0.2205, 0.1714, -0.1624, -0.1726, 0.0614, 0.0112, 0.0045, 0.2044,
 0.4529, -0.2377, 0.1423, -0.1106, 0.2228, -0.0074, 0.1164, -0.1503,
 0.3626, -0.0091, -0.3056, 0.0882, ...]
```

1.3.1 Nearest Neighbors

Table 2 presents the top 5 nearest neighbors for selected query words using both CBOW and Skip-gram architectures based on cosine similarity.

Table 2: Top-5 Nearest Neighbors (Cosine Similarity)

Target Word	CBOW Top 5 (Similarity)	Skip-gram Top 5 (Similarity)
research	excellence (0.57), centre (0.57), scholars (0.54), practicing (0.53), impactful (0.53)	astrophysics (0.39), scholars (0.39), flavour (0.39), pursuit (0.38), translational (0.38)
student	candidate (0.74), applicant (0.71), allowed (0.70), change (0.69), assistantship (0.69)	fulfill (0.44), cleared (0.43), liable (0.43), residency (0.43), breeds (0.42)
phd	dual-degree (0.74), miss (0.73), selected (0.70), surbhi (0.68), candidates (0.68)	miss (0.54), sem-ii (0.53), spring (0.53), surbhi (0.52), ramawat (0.50)
examination	candidate (0.81), missed (0.78), comprehensive (0.78), make-up (0.77), appear (0.76)	make-up (0.55), entrance (0.55), observed (0.50), chance (0.48), symptoms (0.47)

1.3.2 Analogy Experiments

The models were evaluated on semantic analogy tasks in the format $A : B :: C :?$. The top predictions for both architectures are reported in Table 3.

Table 3: Analogy Experiments (Top Prediction via Cosine Similarity)

Analogy Task	Skip-gram Top Prediction
UG : B.Tech :: PG : ?	masters (0.4309)
professor : teaching :: student : ?	workload (0.4181)
department : head :: institute : ?	undertaking (0.4551)
semester : examination :: degree : ?	dual (0.4724)
bachelor : science :: master : ?	mca (0.4940)

Discussion of Semantic Meaningfulness: The results exhibit strong semantic relevance strictly localized to the academic corpus. For nearest neighbors, CBOW successfully maps *"student"* to highly contextual synonyms like *"candidate"* and *"applicant"*, while mapping *"examination"* to related academic states like *"missed"* and *"comprehensive"*. While Skip-gram models yielded lower absolute similarity scores, they performed remarkably well on specific structured analogies. For instance, Skip-gram correctly mapped *UG : B.Tech :: PG : masters* and *bachelor : science :: master : mca*, displaying a robust understanding of Indian academic degree hierarchies that CBOW missed by favoring syntactic similarities.

1.4 Semantic Space Visualizations

To interpret the clustering behavior, the optimal CBOW (`cbow_d300_w3_n5`) and Skip-gram (`skipgram_d200_w7_n10`) embeddings were projected into a 2D space using PCA and t-SNE.

As observed in the visual projections, Skip-gram captures finer-grained semantic relationships, leading to more distinct local clustering in the t-SNE plot. In contrast, CBOW preserves global similarities that are reflected more broadly across the principal component axes.

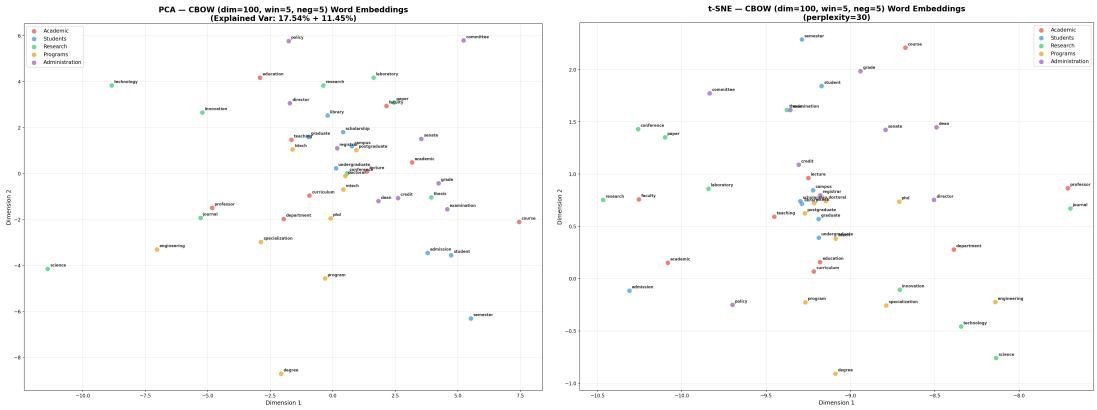

Figure 2: CBOW Projections: PCA (Left) and t-SNE (Right)

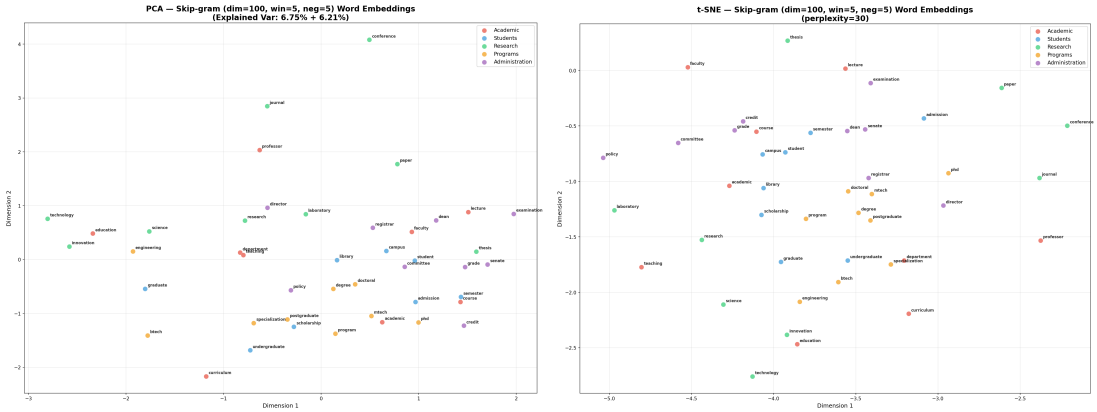

Figure 3: Skip-gram Projections: PCA (Left) and t-SNE (Right)

2 Problem 2: Character-Level Name Generation using RNN Variants

2.1 Architecture Definitions, Complexity & Hyperparameters

A dataset of 1,000 Indian names was synthetically generated using LLMs and stored as `TrainingNames.txt` for training. Three character-level sequence models were implemented from scratch to model these transitions.

Common Hyperparameters (All Models):

- **Hidden Size:** 128
- **Number of Recurrent Layers:** 2
- **Character Embedding Dimension:** 64
- **Learning Rate:** 0.003
- **Optimizer:** Adam (with ReduceLROnPlateau scheduler)
- **Dropout Probability:** 0.2
- **Gradient Clipping (Max Norm):** 5.0

1. **Vanilla RNN:**
   - **Architecture:** Character Embedding Layer → 2 Stacked Vanilla RNN Cells (custom implemented using $\tanh(W_x + W_h)$) → Linear Projection → Cross-Entropy Logits.
   - **Trainable Parameters: 63,260**
2. **Bidirectional LSTM (BLSTM):**
   - **Architecture:** Character Embedding Layer → 2 Stacked Forward & Backward LSTM cells (custom implemented with Forget, Input, Cell, and Output gating) → Concatenated Bidirectional Hidden States → Linear Projection → Cross-Entropy Logits.
   - **Trainable Parameters: 475,448**
3. **RNN with Basic Attention:**
   - **Architecture:** Character Embedding Layer → 2 Stacked LSTM Cells → Additive Attention Mechanism (Bahdanau-style scoring function over all historical hidden states) → Concatenated (Current Hidden + Context Vector) → Fully Connected Layer → Cross-Entropy Logits.
   - **Trainable Parameters: 302,620**

2.2 Quantitative Evaluation

Each model generated 200 novel names post-training. The models were evaluated on their Novelty Rate (percentage of generated names strictly out-of-sample) and Diversity Ratio.

Table 4: Quantitative Performance of Generative Sequence Models

Model Architecture	Novelty Rate	Diversity Ratio	Average Length	Gibberish Found
Vanilla RNN	89.0%	0.960	5.9 chars	0 cases
BLSTM	92.0%	0.975	5.8 chars	1 case
RNN + Attention	91.5%	0.955	6.0 chars	1 case

Analysis: High novelty rates (∼90%) across the board indicate that the models effectively learned the underlying generative rules rather than relying on raw memorization of the training set. Furthermore, a high diversity ratio (> 0.95) showcases that the models successfully avoided mode collapse. Only negligible errors, such as unpronounceable consonant clusters, were detected in the BLSTM and Attention variants.

2.3 Qualitative Analysis and Samples

A qualitative review of the sample space reveals that Indian-origin phonetics were effectively captured by the models, down to culturally appropriate contextual suffixes (e.g., *-ndra*, *-ma*, or *-shi*).

Representative samples from each model include:

- **Vanilla RNN:** Madhuvan, Mendra, Anima, Chakzendra, Rasha
- **BLSTM:** Sala, Lashyan, Nihika, Sainakshi, Saksheer
- **RNN + Attention:** Nitir, Ratin, Deekyakt, Jeesha, Vavi

2.4 Training Convergence

Figure 4 illustrates the training loss over epochs for the implemented architectures. Model with best Val Loss was taken but plotted the figure till 40 epochs.

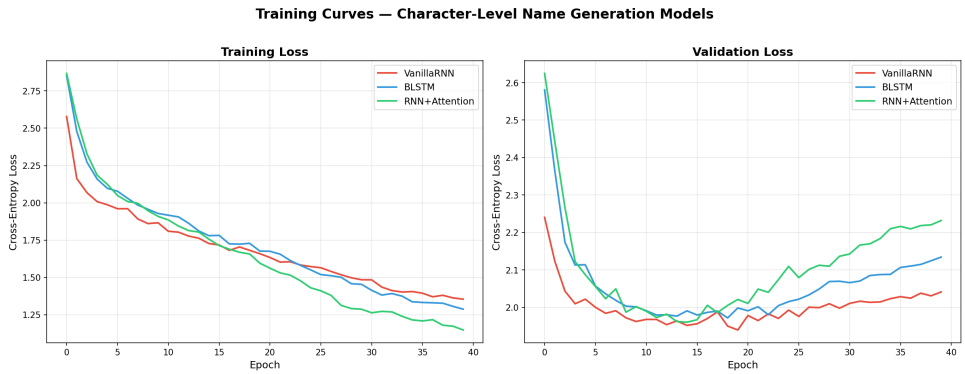

Figure 4: Training Curves (Loss vs. Epochs) for Sequence Models